

CARSON BALL

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Seeking Summer 2027 Computer Engineering Internship

Education

University of Georgia

Athens, GA

Bachelor of Science in Computer Systems Engineering | GPA: 3.42/4.0

Expected May 2028

- **Relevant Coursework:** Design of Digital Systems (Verilog/FPGA), Digital Logic Design, Electronics I, Circuit Analysis, Linear Systems, Systems Programming, Data Structures, Applied Linear Algebra

Technical Skills

Languages: C, C++, ARM Assembly (basic), Python, Verilog, MATLAB, Bash, Java

Embedded: STM32 (ARM Cortex-M), bare-metal & RTOS firmware, embedded C, UART/SPI/I2C, PWM, ADC, interrupts, SWD/ST-Link, IMU, encoders, PID

Hardware: Basys 3 FPGA, Vivado, op-amp/transistor circuits, oscilloscopes

Tools: Git, GitHub, Linux/Unix, GCC, GDB, Valgrind, POSIX threads, Make, Maven, LaTeX

Project Experience

IEEE Micromouse Robot | C/C++, STM32 (ARM Cortex-M), Bare-Metal Firmware

Embedded Systems Competition Team, IEEE UGA Student Branch

Sep 2026 – Present

- Develop bare-metal C/C++ firmware (with some ARM assembly) for an STM32 (ARM Cortex-M) microcontroller on an autonomous maze-solving robot.
- Own the team's flood-fill maze-solving algorithm and implement weighted Dijkstra path planning alongside PID motor control for fast, stable navigation.
- Integrate IR distance sensors, wheel encoders, and an IMU with the microcontroller for wall detection, odometry, and heading correction.
- Configure STM32 peripherals (PWM timers, ADC, I2C, SPI, UART, and timer interrupts) for motor drive and sensor acquisition.

FPGA Digital Design Labs | Verilog, Vivado, Basys 3 (Xilinx Artix-7)

CSEE 4270 – Design of Digital Systems

Aug 2026 – Present

- Designed a binary-to-hexadecimal 7-segment LED decoder in Verilog (RTL) and deployed it to a Basys 3 (Xilinx Artix-7) FPGA.
- Wrote Vivado testbenches to exhaustively simulate all input combinations and verify waveforms before synthesis.
- Edited XDC constraint files to map module I/O to Basys 3 switches and Pmod ports.

Systems Programming Projects | C/C++, Linux

CSCI 1730 – Systems Programming

Jan 2026 – May 2026

- Built a multithreaded POSIX threads program to search grids up to 3,567 x 3,567, comparing 1–3 thread runtimes.
- Wrote a file-comparison tool with low-level Unix system calls, analyzing the speed of byte-by-byte vs. bulk reads.
- Compressed DNA sequences to 2 bits per base with bitwise operators, cutting storage by about 75% versus ASCII.

Work Experience

University of Georgia Libraries

Athens, GA

Senior Security Assistant

May 2025 – Present

- Promoted from Security Assistant to Senior Security Assistant (Dec 2025) based on reliability and performance; train new staff on facility protocols, emergency procedures, and de-escalation.
- Monitor library facilities and resolve patron concerns, escalating issues to supervisors when necessary.

Carrabba's Italian Grill

Athens, GA

Host

Jul 2026 – Present

- Coordinate seating and communicate with guests, servers, and management in a fast-paced team environment.

Jersey Mike's Subs

Bethlehem, GA

Team Member

Jun 2022 – May 2025

- Delivered customer service in a high-volume, fast-paced team environment for three years.

Campus & Community Involvement

IEEE UGA Student Branch

Athens, GA

Member

Aug 2026 – Present

- Selected for the branch's embedded systems competition team, one of several engineering project teams.
- Participate in technical meetings, workshops, and networking events with students and industry professionals.

Honors & Awards

Zell Miller Scholarship, State of Georgia

Aug 2024 – Present

Eagle Scout, Boy Scouts of America

Apr 2021